Spectral Correlation-Based Diverse Band Selection for Hyperspectral Image Classification

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***Abstract*— Band selection (BS), which can reduce the spectral dimensionality effectively, has become one of the most popular topics in hyperspectral image (HSI) analysis. Recently, sparse representation-based BS has emerged as a popular tool. The existing sparse models mainly focus on minimizing reconstruction error and sparsity, while do not fully exploit the unique correlations among hundreds of continuous bands, which may cause representative bands missed and highly correlated bands selected. Therefore, this article proposes the spectral correlation-based diverse BS (SCDBS) for HSIs to improve representativeness and diversity of the selected bands. Specifically, a correlation derived weight is used to perform weighted sparse reconstruction to select the bands that are more correlated with the whole HSI, and a correlation minimization term is designed to remove the highly correlated bands simultaneously. In addition, the proposed method imposes an adjustable sparse constraint by using an $\ell_{2,0<p\leq 1}$ norm, which extends and unifies the commonly used $\ell_{2,1}$ norm to provide more flexible sparsity level. To optimize the proposed BS model, an iteration algorithm with relatively low computational cost is designed, of which the convergence is theoretically presented. Experimental results on three benchmark datasets have demonstrated that the proposed SCDBS outperforms state-of-the-art methods in HSI classification.**

***Index Terms*— Band selection (BS), correlation minimization, hyperspectral images (HSIs), sparse representation.**

I. Introduction

HYPERSPECTRAL sensors can obtain images with hundreds or even thousands of very narrow and continuous spectral bands, which provide favorable conditions for accurate target recognition and classification applications [1]. As a result, hyperspectral images (HSIs) are extensively used in target detection [2], environmental monitoring [3], mineral exploration [4], and agricultural investigation [5].

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Although these large numbers of spectral bands provide more information than traditional RGB channels, they bring challenges to the related image processing techniques. According to the characteristics of hyperspectral imaging, there exists high correlation between adjacent bands. In addition, high-dimensional HSI data not only increase the time and space complexity, but also lead to the Hughes phenomenon [6], which may degrade classification performance. As a result, the dimensionality of HSIs needs to be reduced to avoid the "curse of dimensionality" [7].

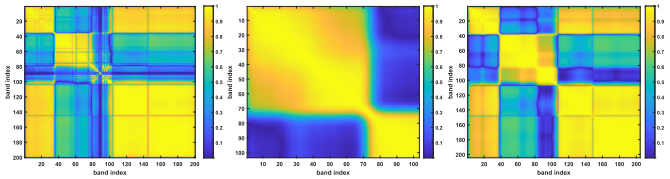
Band selection (BS) is an effective way for dimensionality reduction, which selects a few representative bands from HSIs [8]. To date, numerous BS methods have been put forth and can be generally classified into supervised and unsupervised ones depending on whether training is needed or not. Supervised methods heavily rely on annotated samples, and the bands with strong discriminative information are selected with the help of label information to evaluate class separability. The unsupervised method relies on the data itself to determine how important each band is, and to decide whether to retain a band. Due to the large feature variations of training samples, and the expensive cost of labeling work, supervised methods may be limited. Therefore, unsupervised methods have broader application prospects and have become a hot research direction of BS [9]. Detailed development of BS can be found in the survey [10].

In the unsupervised BS, a ranking-based method [11], [12], [13], which ranks and selects bands according to a certain criterion, a searching-based method [14], [15], [16], which updates the selected bands simultaneously or sequentially through iterations to make the designed criterion function optimal, and a clustering-based method [17], [18], [19], [20], which first groups all bands into different clusters and then selects the band closest to the centroid from each cluster, are the common categories. Recently, benefited from the advances of sparse representation, sparse representation-based methods have been applied for BS [21], [22], through solving an optimization problem with sparsity constraints. In the sparse representation-based BS [10], there are generally two kinds of sparse formulation.

One is sparse subspace clustering (SSC), which regards all bands are sampled from a union of some low-dimensional orthogonal subspaces corresponding to some clusters, and each band can be represented as a linear combination of other bands in the subspace it belongs to [24]. For example, Sun et al. [24] proposed the improved SSC (ISSC), in which

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(a) (b) (c)

Fig. 1. Correlations between spectral bands of three HSIs. (a) Indian Pines. (b) Pavia University. (c) Salinas.

the ℓ_2 norm replaces the ℓ_1 norm in the original SSC [25], and the angular similarity measurement is used to build the similarity matrix. Later, they presented fast and latent low-rank subspace clustering (FLLRSC) for BS [26], where the subspaces are spanned by not only observed but also unobserved bands to provide sufficient sampling bands and to improve robustness to noise, and correntropy measure is proposed to construct the similarity. Zhai et al. [27] argued that the SSC cannot model the global structures of HSIs, and proposed low-rank subspace clustering (LRSC) with a Laplacian regularizer, which can enhance the smoothness of coefficients of adjacent bands to capture the local manifold structure. Recently, Huang et al. [28] performed tensor instead of matrix-based subspace clustering to preserve spatial structure of HSIs and proposed heterogeneous constraints based on 3-D properties to better exploit intrinsic structure.

The other is sparse self-representation (SSR) derived from sparse representation of multimeasurement vectors (MMVs), which regards each band as a linear combination of all bands themselves rather than a union of some low-dimensional subspaces, and enforces a sparsity constraint on the representation coefficient [29]. Based on SSR, Sun et al. [30] proposed dissimilarity-weighted SSR (DWSSR), which adds a dissimilarity-based coding cost regularization term [31] to ensure each band to be represented by a few bands with the smallest dissimilarities. Hu et al. [32] adopted the evolutionary multiobjective optimization to solve the SSR problem. Wei et al. [33] proposed scalable one-pass self-representation learning (SOP-SRL) for big data analysis, which can process sample subsets sequentially. Cai et al. [34] proposed graph convolutional self-representation (GCSR), which integrates graph convolution operation into SSR to use structural information among spectral bands. Zhang et al. [23] proposed marginalized graph self-representation (MGSR), where graph convolution is used to exploit spatial information of homogeneous regions, and infinite corrupted samples are used to build a combination of multiple self-representation to improve robustness.

The existing SSR methods mainly focus on spare reconstruction of data, and some work considers manifolds, but do not fully consider the strong correlation among bands, which is a particularly significant attribute in HSIs, as shown in Fig. 1, where bands, especially the adjacent ones, have very high correlations. As a result, two problems can be caused. The first is that the bands, which have higher correlations with all bands, may be missed in the final selection; however, they may be more representative of the whole HSI. The second is that the highly correlated bands are selected simultaneously, which leads to information redundancy, since if one band can represent the HSI well, the bands with high correlations can also do well. Taking MGSR [23][1] as an example with the default optimal settings, the results on Indian Pines, including the significance of each band and the distribution under different numbers of selected bands, are shown in Fig. 2, from which it can be clearly observed that adjacent bands have almost the same significance, and are selected simultaneously, especially when the number of chosen bands grows. In addition to the above two problems, the existing methods cannot adjust the sparsity level, and the sparsity imposed by the commonly used $\ell_{2,1}$ norm relaxed from $\ell_{2,0}$ is not strong enough, as shown in Fig. 2(a), which also cases the adjacent bands being selected.

Therefore, this article proposes the spectral correlation-based diverse BS (SCDBS) for HSIs to improve both representativeness and diversity of the selected bands. First, SCDBS achieves the representativeness by correlation based on weighted sparse reconstruction, in which each band is given different weights based on the correlation between each band and the whole image, to make the bands more related to the whole HSI be selected. Second, SCDBS achieves diversity by the designed correlation minimization regularization term, through which the two highly correlated bands cannot have large representation coefficients simultaneously; that is, they cannot be selected together. Third, SCDBS adopts the $\ell_{2,0<p\leq 1}$ norm, which incorporates the commonly used $\ell_{2,1}$ norm to implement the adjustable row-sparsity constraint to the coefficient matrix to achieve different requirements of sparsity levels.

The main contributions of this article are summarized as follows.

1) The SCDBS for HSIs is proposed to improve both representativeness and diversity of the selected bands through weighted reconstruction and correlation minimization regularization, respectively.
2) The $\ell_{2,0<p\leq 1}$ norm is integrated to the proposed SCDBS, which unifies the commonly used $\ell_{2,1}$ norm to achieve adjustable constraints on the sparsity level of band representation coefficients.
3) An efficient iterative optimization is designed for SCDBS, and its convergence is theoretically ensured.
4) Extensive comparison and ablation experiments on three benchmark datasets are conducted, which have demonstrated the effectiveness of the proposed method.

In the following, Section II describes the proposed model, Section III presents the model optimization and its analysis, Section IV gives the experiments and discussions, and Section V draws the conclusions.

II. Model Formulation

Given an HSI $\mathcal{X} \in \mathbb{R}^{w\times h\times n}$ capturing $w \times h$ pixels and n spectral bands, the objective of BS is choosing a subset of representative bands. In BS, a 3-D HSI is usually mathematically defined as $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n] \in \mathbb{R}^{m\times n}$, where $\mathbf{x}_i \in \mathbb{R}^m$ is the representation of the ith band and m is the dimension of band representation. Finally, s representative bands specified by $\mathbf{x}_{k_1}, \mathbf{x}_{k_2}, \ldots, \mathbf{x}_{k_s}$ are selected, where $1 \leq k_1 < k_2 < \cdots < k_s \leq n$, and $\mathbf{k} = (k_1, k_2, \ldots, k_s)^T \in \mathbb{R}^s$ is the index vector indicating the selected bands.

[1] https://github.com/ZhangYongshan/MGSR

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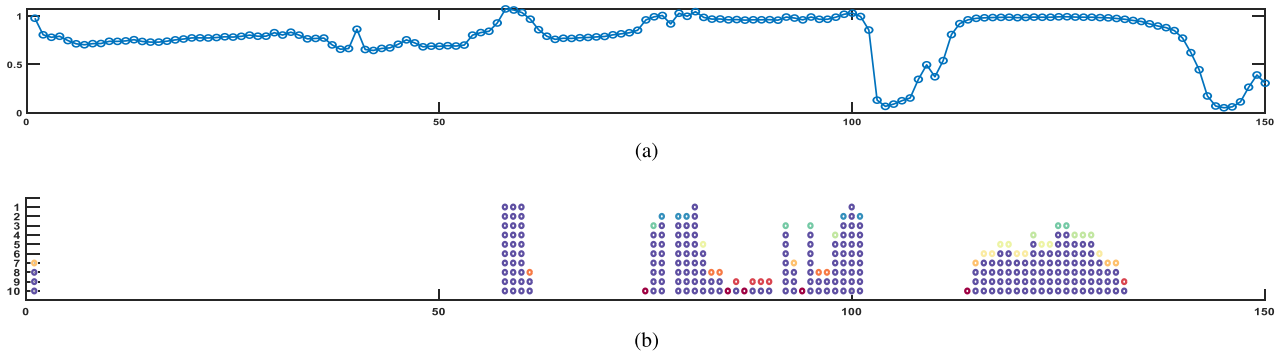


Fig. 2. Illustration of the results of MGSR [23] on the Indian Pines dataset, where only the first 150 bands are presented. (a) *y*-axis is the reconstruction weight of each band computed as ℓ_2 norm of each row of the sparse matrix. (b) *y*-axis represents the selection of 5,10,15, . . . , 50 bands from top to bottom, where the circles marked with a new color in two consecutive rows represent the bands newly selected.

The SSR-based BS commonly introduces an error term and a sparse constraint term, which can be formulated as follows:

$$\min_{\mathbf{W}} \left\|\mathbf{X}-\mathbf{X}\mathbf{W}\right\|_F^2+\alpha\left\|\mathbf{W}\right\|_{2,1} \tag{1}$$

in which the first term measures the reconstruction error, and the second term is the $\ell_{2,1}$ norm constrained row sparsity of the coefficient matrix $\mathbf{W}\in\mathbb{R}^{n\times n}$, computed as follows:

$$\left\|\mathbf{W}\right\|_{2,1}=\sum_{i=1}^{n}\sqrt{\sum_{j=1}^{n}\mathbf{W}_{ij}^2}=\sum_{i=1}^{n}\left\|\mathbf{W}_{i\cdot}\right\| \tag{2}$$

with $\mathbf{W}_{i\cdot}$ being the *i*th row of $\mathbf{W}$, and $\alpha>0$ is a trade-off parameter. Accordingly, the above model promotes $\mathbf{W}$ to have only a few nonzero rows with the row-sparsity constrained by $\ell_{2,1}$ norm.

In HSIs, each pixel contains hundreds of bands, forming a complete continuous spectral curve, which reflects the characteristics of the pixel. Due to the small difference of the central wavelength of the adjacent bands, their correlation is very strong. However, this unique and important attribute has not been fully explored by the existing methods. Therefore, this article will use intercorrelation to carry out SSR-based BS, and the two important attributes of the final selected bands, namely, representativeness and diversity, will be taken into account. In other words, we expect that the bands with stronger correlations with all bands can be selected; meanwhile, the correlations among the selected bands can be reduced.

In general, the smaller the dissimilarity between a pair of data, the more fully the data can represent each other [31]. Similarly, if some band has a higher correlation with all bands in an HSI, it should be more representative of the whole HSI. As a result, it should be more likely to be selected. Since the bands are selected according to the each row of $\mathbf{W}$, each row can be assigned a weight to affect the final selection, which is formulated as follows:

$$\begin{bmatrix}\mathbf{D}_{11}\mathbf{W}_{1\cdot}\\ \vdots\\ \mathbf{D}_{nn}\mathbf{W}_{n\cdot}\end{bmatrix}=\begin{bmatrix}\mathbf{D}_{11} & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & \cdots & \mathbf{D}_{nn}\end{bmatrix}\begin{bmatrix}\mathbf{W}_{1\cdot}\\ \vdots\\ \mathbf{W}_{n\cdot}\end{bmatrix}=\mathbf{D}\mathbf{W} \tag{3}$$

where $\mathbf{D}_{ii}$ is the weight of the *i*th band.

Therefore, the formulation for correlation-based weighted SSR for BS is

$$\min_{\mathbf{W}} \left\|\mathbf{X}-\mathbf{X}\mathbf{W}\right\|_F^2+\alpha\left\|\mathbf{D}\mathbf{W}\right\|_{2,1} \tag{4}$$

and the weight for each band is computed based on correlations as follows:

$$\mathbf{D}_{ii}=1-\mathcal{N}\left(\sum_{j=1}^{n}\mathbf{C}_{ij}\right) \tag{5}$$

where $\mathbf{C}_{ij}$ is the correlation between the *i*th and *j*th bands, and $\mathcal{N}(\cdot)$ is a normalization function with a range of $[0,1]$. It can be observed from (4) that a smaller $\mathbf{D}_{ii}$ will cause greater values of $\mathbf{W}_{i\cdot}$; then, the significance of the *i*th band will be increased. Based on (5), if a certain band has stronger correlation with the whole HSI, it will be more likely to be selected. Therefore, representativeness of selected bands is improved.

In HSIs, the bands with high correlations provide redundancy and cannot be chosen together so as to improve the diversity of the final selection. In this article, the following term is designed to minimize the interband correlation:

$$\min_{\mathbf{W}_{\cdot j}} \mathbf{W}_{\cdot j}^T\mathbf{C}\mathbf{W}_{\cdot j} \tag{6}$$

in which $\mathbf{W}_{\cdot j}$ is the *j*th row of $\mathbf{W}$, denoting the coefficient of the *j*th band being represented by all bands.

If the *k*th and the *l*th band are highly correlated, $\mathbf{C}_{k,l}$ is large. To minimize the regularization term in (6), the *k*th and the *l*th element of $\mathbf{W}_{\cdot j}$ will not be great at the same time. As a result, only one band alone is primarily responsible for reconstructing the *j*th band. When *n* bands are modeled together, the above minimization term for the *j*th band is modified as the sum of *n* bands

$$\min_{\mathbf{W}_{\cdot j},1\le j\le n}\sum_{j=1}^{n}\mathbf{W}_{\cdot j}^T\mathbf{C}\mathbf{W}_{\cdot j}=\min_{\mathbf{W}} Tr\left(\mathbf{W}^T\mathbf{C}\mathbf{W}\right). \tag{7}$$

Compared with (6), (7) enforces that the *k*th and *l*th rows of $\mathbf{W}$ cannot be large simultaneously. As a result, only one of the highly correlated bands will have a high significance and be selected. Thus, diversity of selected bands is improved. Then, the model in (4) can be updated as follows:

$$\min_{\mathbf{W}} \left\|\mathbf{X}-\mathbf{X}\mathbf{W}\right\|_F^2+\alpha\left\|\mathbf{D}\mathbf{W}\right\|_{2,1}+\beta Tr\left(\mathbf{W}^T\mathbf{C}\mathbf{W}\right) \tag{8}$$

in which $\alpha>0$ and $\beta>0$ are both trade-off parameters.

The model in (8) takes the representativeness and diversity of selected bands into consideration through interband correlation; however, the $\ell_{2,1}$ norm has disadvantages. The first

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is that the sparsity level cannot be adjusted, while a sparsity level from "very sparse" to "not very sparse" may be required in practice. The second is that the $\ell_{2,1}$ norm relaxed from the $\ell_{2,0}$ norm is not sparse enough and may lead to similar representation coefficients in adjacent rows and the selection of adjacent bands. Thus, this article proposes to use the adjustable $\ell_{2,0<p\leq 1}$ norm to constrain the row sparsity, which is able to adjust the sparsity level through p. As a result, the proposed SCDBS for HSIs can be finally formulated as follows:

$$\min_{\mathbf{W}} \left\|\mathbf{X}-\mathbf{X}\mathbf{W}\right\|_F^2+\alpha\left\|\mathbf{D}\mathbf{W}\right\|_{2,p}+\beta Tr\left(\mathbf{W}^T\mathbf{C}\mathbf{W}\right) \quad (9)$$

where the $\ell_{2,p}$ norm of a matrix is computed as follows:

$$\left\|\mathbf{W}\right\|_{2,p}=\sum_{i=1}^{n}\left(\sqrt{\sum_{j=1}^{n}\mathbf{W}_{ij}^2}\right)^p=\sum_{i=1}^{n}\left\|\mathbf{W}_{i\cdot}\right\|_2^p. \quad (10)$$

When $0<p<1$, $\ell_{2,p}$ can better approximate the $\ell_{2,0}$ norm; when $p=1$, $\ell_{2,p}$ becomes the commonly used $\ell_{2,1}$ norm.

In (9), the first term is the common representation error, which measures the ability of the selected bands to cover the original HSI information; the second one is the weighted and adjustable sparsity, that makes the row sparsity related to the correlation through the enforced weights, and meets different sparsity levels through the configuration of p; the third one is the correlation minimization regularization term, which reduces the possibility of highly correlated bands being selected together.

To select a band subset, the contribution curve is formed by computing the ℓ_2 norm of each row of $\mathbf{W}$; then, the curve is divided into s segments, and the band having the maximum contribution in each segment is selected. In this article, the number of selected bands can be any integer given, which facilitates the performance exploration and comparison by flexibly setting the different numbers of selected bands. In practical applications, the recommendation method of an appropriate band number has been widely studied and can also be applied to our method, such as isolated-point-stopping criterion in [19] and determinantal point process (DPP)-based method in [35]. However, different methods generally give inconsistent estimation results. For example, the proper number of selected bands for the Indian Pines dataset is 10 in E-FDPC [19], 14 in TRC-OC-FDPC [20], 6 in fast neighborhood grouping for band selection (FNGBS) [35], and 27 in [36]. Therefore, it is also necessary to carefully select the estimation method.

III. Model Optimization

In this section, the optimization of the proposed SCDBS is first presented; then, the convergence and computational complexity are theoretically analyzed.

A. Iteration Optimization Algorithm

Since the issue in (9) is not smooth, it is difficult to directly get its solution. Following the optimization of the $\ell_{2,1}$ norm-based problems [37], we design an iterative optimization to solve it efficiently.

Suppose that $\mathbf{W}_{t-1}$ is obtained after the $(t-1)$th iteration. In the tth iteration, the problem in (9) can be transformed to

$$\min_{\mathbf{W}} \left\|\mathbf{X}-\mathbf{X}\mathbf{W}\right\|_F^2+\alpha Tr\left(\mathbf{W}^T\mathbf{D}^T\mathbf{P}_t\mathbf{D}\mathbf{W}\right)+\beta Tr\left(\mathbf{W}^T\mathbf{C}\mathbf{W}\right) \quad (11)$$

in which $\mathbf{P}_t\in\mathbb{R}^{n\times n}$ is a diagonal matrix, and $(\mathbf{P}_t)_{ii}$ can be calculated as follows:

$$(\mathbf{P}_t)_{ii}=\frac{p}{2\left\|\left(\mathbf{D}\mathbf{W}_{t-1}\right)_{i\cdot}\right\|_2^{2-p}} \quad (12)$$

in which $(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}$ is the ith row of $\mathbf{D}\mathbf{W}_{t-1}$.

In the iteration, $\mathbf{P}_t$ is obtained according to (12), and $\mathbf{X}$, $\mathbf{C}$, and $\mathbf{D}$ are all known. Moreover, the problem of (11) is convex with respect to $\mathbf{W}$. Therefore, the derivative of the objective function to $\mathbf{W}$ can be set as zero to find the solution of (11), which is formulated as follows:

$$\mathbf{X}^T\mathbf{X}\mathbf{W}-\mathbf{X}^T\mathbf{X}+\alpha\mathbf{D}\mathbf{P}_t\mathbf{D}\mathbf{W}+\beta\mathbf{C}\mathbf{W}=0. \quad (13)$$

Thus, $\mathbf{W}_t$ can be obtained as follows:

$$\mathbf{W}_t=\left(\mathbf{X}^T\mathbf{X}+\alpha\mathbf{D}\mathbf{P}_t\mathbf{D}+\beta\mathbf{C}\right)^{-1}\mathbf{X}^T\mathbf{X}. \quad (14)$$

With the increase in iterations, the value of the objective function in (9) will increasingly get closer to its minimum. In other words, the error of two successive objective values will become small, which is defined in (15), and can be used as an indicator to stop iterations. In general, the maximum number of iterations T_{max} is another indicator. As a result, the iteration will stop whenever ∇_t is below a predetermined threshold ∇_{Th} or when t is greater than T_{max}. In this article, T_{max} and ∇_t are set to 50 and 10^{-5}, which is based on the practice of similar optimization algorithms. Of course, the thresholds can also be adjusted according to the dataset to be carried out

$$\nabla_t=\begin{cases}O_0, & t=0\\ |O_t-O_{t-1}|/O_{t-1}, & t\geq 1\end{cases} \quad (15)$$

where O_t is the objective function value at the tth iteration.

The overall optimization is presented in Algorithm 1.

Algorithm 1 Optimization Algorithm of SCDBS

Input: a hyperspectral image with spatial transformation $\mathbf{X}\in\mathbb{R}^{m\times n}$, a sparsity level control parameter p, regularization term balance parameters α and β.

Output: sparse representation coefficient matrix $\mathbf{W}$.

Initialization: $t=0$, randomizing $\mathbf{W}_0$, calculate ∇_0.

Compute the spectral correlation matrix $\mathbf{C}\in\mathbb{R}^{n\times n}$ and the diagonal weight matrix $\mathbf{D}\in\mathbb{R}^{n\times n}$.
while $t\leq T_{max}$ and $\nabla_t\geq\nabla_{Th}$ **do**
  Increasing $t\leftarrow t+1$.
  Computing $\mathbf{P}_t$ through (12).
  Computing $\mathbf{W}_t$ through (14).
  Computing ∇_t through (15).
end while

B. Convergence Proof

Prior to demonstrating the convergence, we first provide two lemmas.

Lemma 1: When $0<p\leq 1$, $px^2/2-x^p+1-p/2\geq 0$ holds for any scalar $x>0$.

Proof: After constructing the function $f(x)=px^2/2-x^p+1-p/2$, the first-order derivative is computed as follows:

$$f'(x)=px-px^{p-1} \quad (16)$$

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and the second-order derivative is

$$f''(x) = p - p(p-1)x^{p-2}. \tag{17}$$

Obviously, for $x > 0$, $f''(x) > 0$, indicating that $f'(x)$ is monotonically increasing. Then, it is found that $f'(1) = 0$. Therefore, when $x < 1$, $f'(x) < 0$, and when $x > 1$, $f'(x) > 0$. It is indicated that $f(x)$ first decreases and then increases, and $f(1)$ is its global minimum. Therefore

$$f(x) = px^2/2 - x^p + 1 - p/2 \geq f(1) = 0. \tag{18}$$

Lemma 2: When $0 < p \leq 1$, the following inequality holds for any nonzero vectors $\mathbf{u}$, $\mathbf{v}$:

$$\|\mathbf{u}\|_2^p - \frac{p\|\mathbf{u}\|_2^2}{2\|\mathbf{v}\|_2^{2-p}} \leq \|\mathbf{v}\|_2^p - \frac{p\|\mathbf{v}\|_2^2}{2\|\mathbf{v}\|_2^{2-p}}. \tag{19}$$

Proof: Let $x = (\|\mathbf{u}\|_2/\|\mathbf{v}\|_2)$, and substitute x into Lemma 1

$$\begin{aligned} &\frac{p}{2}\left(\frac{\|\mathbf{u}\|_2}{\|\mathbf{v}\|_2}\right)^2 - \left(\frac{\|\mathbf{u}\|_2}{\|\mathbf{v}\|_2}\right)^p + 1 - \frac{p}{2} \geq 0 \\ \Rightarrow\ &\frac{p\|\mathbf{u}\|_2^2}{2\|\mathbf{v}\|_2^2} - \frac{\|\mathbf{u}\|_2^p}{\|\mathbf{v}\|_2^p} + \frac{\|\mathbf{v}\|_2^p}{\|\mathbf{v}\|_2^p} - \frac{p\|\mathbf{v}\|_2^2}{2\|\mathbf{v}\|_2^2} \geq 0 \\ \Rightarrow\ &\frac{p\|\mathbf{u}\|_2^2}{2\|\mathbf{v}\|_2^{2-p}} - \|\mathbf{u}\|_2^p + \|\mathbf{v}\|_2^p - \frac{p\|\mathbf{v}\|_2^2}{2\|\mathbf{v}\|_2^{2-p}} \geq 0. \end{aligned} \tag{20}$$

Theorem 1:
The objective function defined in (9) is monotonically decreased in each iteration by using Algorithm 1.

Proof: Let $\mathcal{J}(\mathbf{W}) = \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \beta Tr(\mathbf{W}^T\mathbf{C}\mathbf{W})$ represent the terms other than $\ell_{2,p}$ norm, and the objective function defined by (9) can be rewritten as follows:

$$\min_{\mathbf{W}} \mathcal{J}(\mathbf{W}) + \alpha\|\mathbf{D}\mathbf{W}\|_{2,p}. \tag{21}$$

Similarly, the problem in (11) to be solved in each iteration can be rewritten as follows:

$$\min_{\mathbf{W}} \mathcal{J}(\mathbf{W}) + \alpha Tr\left(\mathbf{W}^T\mathbf{D}^T\mathbf{P}_t\mathbf{D}\mathbf{W}\right). \tag{22}$$

At the tth iteration, $\mathbf{P}_t$ computed by (12) has been known, and $\mathbf{W}_t$ obtained by (14) is the optimal solution to the problem of (11) and (22). In other words, at the tth iteration, $\mathbf{W}_t$ can minimize the objective function in (22) more than $\mathbf{W}_{t-1}$, which can be formulated as follows:

$$\begin{aligned} &\mathcal{J}(\mathbf{W}_t) + \alpha Tr\left(\mathbf{W}_t^T\mathbf{D}^T\mathbf{P}_t\mathbf{D}\mathbf{W}_t\right) \\ &\leq \mathcal{J}(\mathbf{W}_{t-1}) + \alpha Tr\left(\mathbf{W}_{t-1}^T\mathbf{D}^T\mathbf{P}_t\mathbf{D}\mathbf{W}_{t-1}\right). \end{aligned} \tag{23}$$

The calculation of $\mathbf{P}_t$ in (12) results in

$$Tr\left(\mathbf{W}_t^T\mathbf{D}^T\mathbf{P}_t\mathbf{D}\mathbf{W}_t\right) = \sum_{i=1}^{n} \frac{p\left\|(\mathbf{D}\mathbf{W}_t)_{i\cdot}\right\|_2^2}{2\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^{2-p}} \tag{24}$$

$$Tr\left(\mathbf{W}_{t-1}^T\mathbf{D}^T\mathbf{P}_t\mathbf{D}\mathbf{W}_{t-1}\right) = \sum_{i=1}^{n} \frac{p\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^2}{2\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^{2-p}}. \tag{25}$$

Substituting (24) and (25) to inequality (23) yields

$$\begin{aligned} &\mathcal{J}(\mathbf{W}_t) + \alpha\sum_{i=1}^{n} \frac{p\left\|(\mathbf{D}\mathbf{W}_t)_{i\cdot}\right\|_2^2}{2\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^{2-p}} \\ &\leq \mathcal{J}(\mathbf{W}_{t-1}) + \alpha\sum_{i=1}^{n} \frac{p\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^2}{2\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^{2-p}}. \end{aligned} \tag{26}$$

TABLE I
COMPUTATIONAL COMPLEXITY OF THE PROPOSED METHOD

Calculation	Complexity
initializing $\mathbf{C}$ and $\mathbf{D}$	$\mathcal{O}(mn^2)$
updating $\mathbf{P}_t$	$\mathcal{O}(n^2)$
updating $\mathbf{W}_t$	$\max\left\{\mathcal{O}(mn^2), \mathcal{O}(n^3)\right\}$
updating ∇_t	$\mathcal{O}(n^2)$
Whole Algorithm	$\mathcal{O}\left(mn^2T_{max}\right)$

Set the vectors $\mathbf{u}$ and $\mathbf{v}$ in Lemma 2 as $(\mathbf{D}\mathbf{W}_t)_{i\cdot}$ and $(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}$, respectively, and the following inequality is deduced:

$$\begin{aligned} &\alpha\sum_{i=1}^{n}\left\|(\mathbf{D}\mathbf{W}_t)_{i\cdot}\right\|_2^p - \alpha\sum_{i=1}^{n}\frac{p\left\|(\mathbf{D}\mathbf{W}_t)_{i\cdot}\right\|_2^2}{2\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^{2-p}} \\ &\leq \alpha\sum_{i=1}^{n}\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^p - \alpha\sum_{i=1}^{n}\frac{p\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^2}{2\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^{2-p}}. \end{aligned} \tag{27}$$

Combining two sides of inequalities (26) and (27) results in

$$\begin{aligned} &\mathcal{J}(\mathbf{W}_t) + \alpha\sum_{i=1}^{n}\left\|(\mathbf{D}\mathbf{W}_t)_{i\cdot}\right\|_2^p \leq \mathcal{J}(\mathbf{W}_{t-1}) \\ &\quad + \alpha\sum_{i=1}^{n}\left\|(\mathbf{D}\mathbf{W}_{t-1})_{i\cdot}\right\|_2^p \end{aligned} \tag{28}$$

which is

$$\mathcal{J}(\mathbf{W}_t) + \alpha\left\|(\mathbf{D}\mathbf{W}_t)\right\|_{2,p} \leq \mathcal{J}(\mathbf{W}_{t-1}) + \alpha\left\|(\mathbf{D}\mathbf{W}_{t-1})\right\|_{2,p}. \tag{29}$$

Thus, Theorem 1 is proved. In addition, the problem $\mathcal{J}(\mathbf{W})$ in (21) is convex, and the iterative method will converge to a global minimum.

C. Computational Complexity

The computational complexity of the proposed method is analyzed and shown in Table I. Since the dimension of the band vector is usually greater than the number of bands in an HSI, that is, $m > n$, the overall computational complexity of the proposed method can be estimated approximately as $\mathcal{O}(mn^2T_{max})$, which is acceptable when compared with the existing work.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

1) Datasets: Three real-world HSI datasets commonly used in the existing work are adopted, which are detailed as follows.

Indian Pines: It was acquired by the AVIRIS sensor over the Indian Pines test site in the Northwestern Indiana in 1992. The dataset has 145×145 pixels with a spatial resolution of 20 m and 224 spectral bands with a spectral resolution of 10 nm. Finally, 200 bands are remained after removing 20 water absorption bands [104–108], [150–163], and [220] [38].

Pavia University: It was gathered by the ROSIS sensor over the Pavia University, Italy. The dataset has 610×340 pixels

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with a spatial resolution of 1.3 m and 115 bands in the range of 0.43–0.86 μm. In this dataset, 103 bands are remained after removing 12 noisy bands [38].

Salinas: It was gathered by the AVIRIS sensor over Salinas Valley, CA. The dataset has 512×217 pixels with a spatial resolution of 3.7 m and 224 bands with wavelengths varying from 400 to 2500 nm. Finally, 204 bands are retained after removing 20 bands with water absorption and low SNR.

2) Comparison Methods: The proposed SCDBS will be compared with the following representative methods.

1) Maximum variance principal component analysis (MVPCA) [8]: It assesses band prioritization and ranks all bands by constructing a covariance matrix.
2) TRC-OC-FDPC [20]: It uses the optimal clustering framework (OCF). First, a TRC-based objective function is created. Then, OCF is used to determine the ideal band division. Next, the significance of each band is assessed by using E-FDPC.
3) FNGBS [35]: It spectrally partitions an HSI into groups by using coarse-fine strategy to mine the context information in a large spectrum range and obtains the most relevant and informative bands based on local density and information entropy.
4) SimRank [39]: It builds a similarity matrix for HSIs by customizing SSIM [40] to take spatial information into account and then ranks all the bands based on the average similarity and the significant dissimilarity of the bands.
5) SSR [29]: It first finds the indices of nonzero rows of the coefficient matrix after solving SSR with alternating direction method of multipliers (ADMM) and then uses k-means to cluster the nonzero row vectors to select those closest to the centroid within each cluster to form the final band subset.
6) MDSR [41]: It builds multidictionary sparse representation (MDSR), which obtains the sparse solution for each band and the corresponding dictionary and derives the contribution of each band by computing the histogram of nonzero entries.
7) GCSR [34]: It integrates graph convolution into SSR to consider the relationship between bands, and the obtained sparse coefficient matrix is used to conduct spectral clustering, and the band closest to cluster centroids are selected as representative bands.

3) Parameter Setting: Among the above compared methods, MVPCA, TRC-OC-FDPC, FNGBS, and SimRank are free of parameter. For SSR, the parameter λ is set as 15, 20, and 30 through grid search for Indian Pines, Pavia University, and Salinas, respectively. For MDSR, its parameter N controlling the sampled pixels and K_0 controlling the sparsity are set as 50 and 6 for all dataset according to the original paper [41]. For GCSR, its parameter k for k-nearest neighborhood (KNN) and λ are set as 5 and 10^4 for all datasets according to the original paper [34]. Using grid search, α and β of the proposed SCDBS are set as 10 and 100, 100 and 100, and 10 and 100 for three datasets, respectively. In addition, p is fixed as 0.5 except the experiments, which explore the influence of p. Finally, the hyperparameter settings for the proposed and the compared methods are listed in Table II.

4) Classification Settings: The performance of the compared BS methods is assessed by classification of the selected bands. Classification performance is evaluated by two classifiers, including KNN and support vector machine (SVM). In these experiments, 10% of the samples from each class are randomly selected for training, and the remaining ones are utilized for testing. In order to reduce the effect of randomness of training samples, the finally presented result is achieved by averaging ten individual tests. To assess the classification performance, overall accuracy (OA) is used as the measure of accuracy, which is defined as follows:

$$\mathrm{OA} = \frac{\text{number of correctly classified samples}}{\text{number of all test samples}}. \tag{30}$$

TABLE II
PARAMETER SETTINGS OF COMPARED BS METHODS

	Indian Pines	Pavia University	Salinas
MVPCA	—		
TRC-OC-FDPC			
FNGBS			
SimRank			
SSR	$\lambda = 15$	$\lambda = 20$	$\lambda = 30$
MDSR	$N = 50, K_0 = 6$		
GCSR	$k = 5, \lambda = 10^4$		
SCDBS	$\alpha = 10$	$\alpha = 100$	$\alpha = 10$
	$\beta = 100$		

B. Comparisons of Classification

The experiment is used to evaluate the classification performance by setting different sizes of selected bands. Specifically, the size of the band subset varies from 5 to 50 with a step of 5. On each of the used datasets, two groups of results will be compared, including the following: 1) OA curves, i.e., OA values with respect to the number of selected bands, and 2) average OA of all the different numbers of bands.

1) OA Under Different Numbers of Bands: The classification results of different BS methods with KNN and SVM classifiers on three datasets are shown in Figs. 3 and 4, respectively. From these figures, the following important observations can be obtained.

1) The proposed SCDBS outperforms other compared methods in almost all cases in terms of OA for both SVM and KNN classifiers, which indicates the superiority and effectiveness of the proposed SCDBS by modeling and integrating the correlations among spectral bands into sparse presentation. Specifically, the proposed method shows obvious advantages on Indian Pines and Pavia University datasets, and it also slightly outperforms or is comparable to state-of-the-art methods. In addition, the performance evolutions of KNN and SVM in OA are relatively in consistency with each other, while SVM always can achieve higher accuracies compared with KNN using the same bands, which is also consistent with their generally recognized performance.
2) For all the methods on each dataset evaluated by each classifier, the OA is low when the band number is less than 10, and it then rises dramatically with the increase in selected bands. After a certain threshold of the band number, the OA changes slowly, and most curves become almost flat with few fluctuations or even descend on some datasets. The reason may be that the increased

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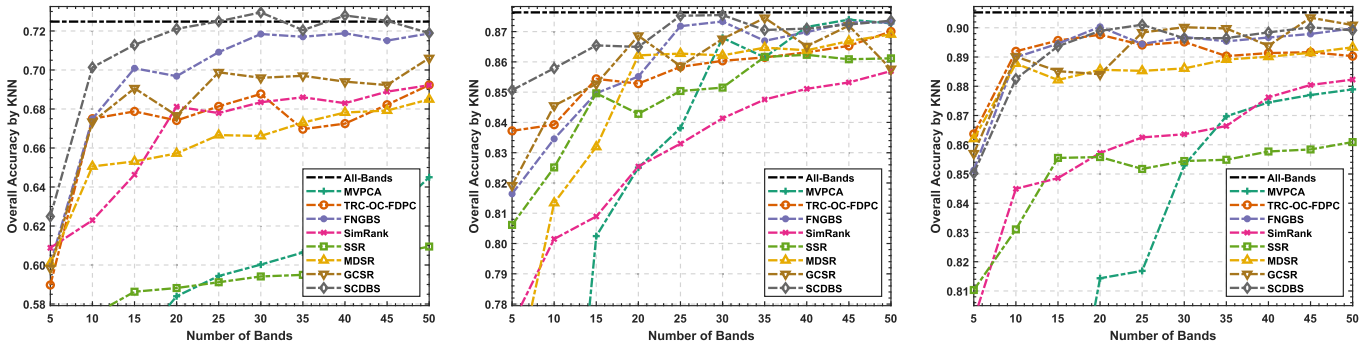


Fig. 3. OA curves by KNN classifier under different numbers of selected bands. OA curves for (a) Indian Pines, (b) Pavia University, and (c) Salinas.

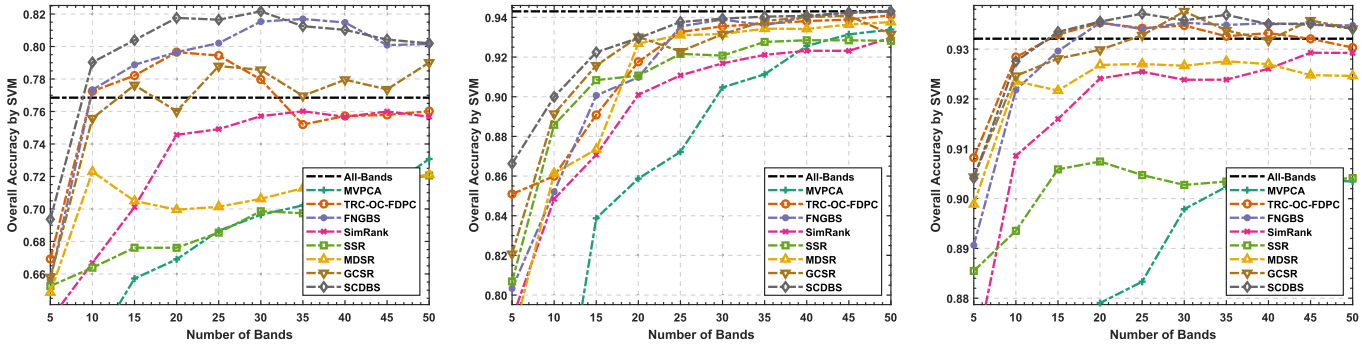


Fig. 4. OA curves by SVM classifier under different numbers of selected bands. OA curves for (a) Indian Pines, (b) Pavia University, and (c) Salinas.

spectral bands can lead to more distinguishable spectral signatures of ground objects, while too many bands will make information redundant. In addition, it is found that when the number of selected bands is sufficiently large but much less than that of all bands in the original HSIs, the classification performance of most BS methods is similar to or even better than that from using all bands, which proves the feasibility and necessity of BS for HSIs through selecting informative bands and removing redundancy.

3) In the ranking-based methods, MVPCA has much lower OA than others in most cases, especially when the number of selected bands is small. It conducts selection according to variance of bands, and the larger variance can make more informative bands be selected sometimes, while also may lead to the selection of noisy bands, which usually present moderate variance. For another ranking-based method SimRank, the relationship among bands reflected by the designed similarity and dissimilarity criteria is taken into consideration; thereby, its performance is improved. While the modeling at the imaging data level is completely ignored, some representative bands are missed.
4) In the sparse representation-based methods, SSR, which is a basic method involving sparse representation, does not perform very well on some datasets, which may be caused by the fact that it only adopts SSR, while does not consider other attributes in HSIs. MDSR adopts multidictionaries to find the sparse coefficient for each band, which may make the representation more accurate. Consequently, it can improve the classification performance compared with SSR. However, it only samples dozens of pixels for each band from the original tens of thousands of pixels, which may lead to the loss of meaningful physical information, thus affecting BS and subsequent classification. GCSR uses graph convolution to consider the relationship among bands and achieves satisfactory classification results.
5) As a clustering-based method, TRC-OC-FDPC obtains higher accuracy than the compared ranking-based methods, which demonstrates the effectiveness of clustering for BS. However, it assumes that the bands in the same cluster should be contiguous. Thus, it may assign the nonadjacent but highly similar bands into different clusters. Accordingly, it may lead to redundancy of selected bands, which degrades the classification performance, especially when the number of desired bands is large.
6) FNGBS performs better than most methods, which may be due to that its coarse-fine mechanism for partitioning the cube into groups, and local selection strategy based on local density and information entropy can well exploit global context information and local distribution, respectively. In addition, it does not change the original information in the HSI cube through spatial transformation used in other methods, which can maintain some key spatial information useful for classification.

2) Average OA and Kappa of Different Numbers of Bands: The average OA values over all cases of different BS methods with KNN and SVM classifiers on three datasets are listed in

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TABLE III
AVERAGED OA BY KNN OVER DIFFERENT NUMBERS (5:5:50) OF SELECTED BANDS (SHOWN AS MEAN% ± STD%)

	All Bands	MVPCA	TRC-OC-FDPC	FNGBS	SimRank	SSR	MDSR	GCSR	SCDBS
Indian Pines	72.4808 ±0.0000	57.9591 ±6.0091	67.0333 ±2.9157	69.7028 ±3.6680	66.7063 ±2.9961	59.2036 ±1.1989	66.0986 ±2.4044	68.2322 ±3.1091	**71.0720** ±3.1255
Pavia University	87.6342 ±0.0000	80.5451 ±10.1797	85.6233 ±1.0719	85.8379 ±1.9585	82.9257 ±2.7114	84.7161 ±1.8393	84.4650 ±3.7931	85.8153 ±1.6377	**86.7710** ±0.8110
Salinas	90.5256 ±0.0000	79.6084 ±10.8463	89.0178 ±0.9622	**89.1706** ±1.4510	85.8243 ±2.3894	84.9069 ±1.5925	88.5292 ±0.8810	89.1297 ±1.3790	89.1696 ±1.5498

TABLE IV
AVERAGED OA BY SVM OVER DIFFERENT NUMBERS (5:5:50) OF SELECTED BANDS (SHOWN AS MEAN% ± STD%)

	All Bands	MVPCA	TRC-OC-FDPC	FNGBS	SimRank	SSR	MDSR	GCSR	SCDBS
Indian Pines	76.8508 ±0.0000	66.9622 ±6.2594	76.2176 ±3.6249	78.6643 ±4.7699	72.8566 ±4.5895	68.9982 ±2.3038	70.4868 ±2.1343	76.3669 ±3.8804	**79.7266** ±3.7539
Pavia University	94.3049 ±0.0000	85.3410 ±9.8412	91.4243 ±3.4488	91.0167 ±4.7169	89.3309 ±4.5220	90.6688 ±3.7548	90.4372 ±5.2660	91.6307 ±3.6665	**92.6191** ±2.4899
Salinas	93.2096 ±0.0000	87.1945 ±4.3760	93.0196 ±0.7996	92.8639 ±1.4014	91.7332 ±1.8804	90.1392 ±0.6696	92.2856 ±0.8640	92.9272 ±0.9524	**93.1471** ±0.9989

TABLE V
AVERAGED KAPPA BY KNN OVER DIFFERENT NUMBERS (5:5:50) OF SELECTED BANDS (SHOWN AS MEAN ± STD)

	All Bands	MVPCA	TRC-OC-FDPC	FNGBS	SimRank	SSR	MDSR	GCSR	SCDBS
Indian Pines	0.7024 ±0.0000	0.5557 ±0.0598	0.6457 ±0.0289	0.6738 ±0.0365	0.6458 ±0.0312	0.5621 ±0.0292	0.6394 ±0.0250	0.6597 ±0.0328	**0.6889** ±0.0317
Pavia University	0.8432 ±0.0000	0.7607 ±0.1132	0.8195 ±0.0126	0.8221 ±0.0232	0.7869 ±0.0320	0.8066 ±0.0134	0.8045 ±0.0448	0.8221 ±0.0197	**0.8321** ±0.0097
Salinas	0.8973 ±0.0000	0.7823 ±0.1136	0.8815 ±0.0104	0.8827 ±0.0159	0.8474 ±0.0253	0.8321 ±0.0250	0.8762 ±0.0092	**0.8832** ±0.0148	0.8826 ±0.0165

TABLE VI
AVERAGED KAPPA BY SVM OVER DIFFERENT NUMBERS (5:5:50) OF SELECTED BANDS (SHOWN AS MEAN ± STD)

	All Bands	MVPCA	TRC-OC-FDPC	FNGBS	SimRank	SSR	MDSR	GCSR	SCDBS
Indian Pines	0.7477 ±0.0000	0.6450 ±0.0684	0.7396 ±0.0390	0.7647 ±0.0498	0.7060 ±0.0489	0.6613 ±0.0427	0.6818 ±0.0240	0.7411 ±0.0390	**0.7788** ±0.0400
Pavia University	0.9259 ±0.0000	0.8160 ±0.1200	0.8896 ±0.0425	0.8849 ±0.0586	0.8634 ±0.0572	0.8784 ±0.0424	0.8779 ±0.0663	0.8930 ±0.0470	**0.9048** ±0.0313
Salinas	0.9260 ±0.0000	0.8617 ±0.0468	0.9238 ±0.0083	0.9221 ±0.0149	0.9107 ±0.0201	0.8916 ±0.0093	0.9156 ±0.0089	0.9232 ±0.0102	**0.9252** ±0.0108

Tables III and IV, where the best and second best results are in bold and underlined, respectively.

From Tables III and IV, the same conclusion as that in Figs. 3 and 4, which is that the proposed method outperforms the other compared methods. Specifically, on Indian Pines and Pavia University, the proposed method greatly improves the average OA of KNN by 1.37% and 0.93%, respectively, and also improves that of SVM by 1.06% and 0.99%, respectively. On the Salinas, the proposed method does not present obvious advantages as on the other two datasets. When KNN is used for classification, our method and FNGBS achieve almost the same highest average OA 89.17%; when SVM is used, our method improves the average OA by 0.13% compared with the second best TRC-OC-FDPC.

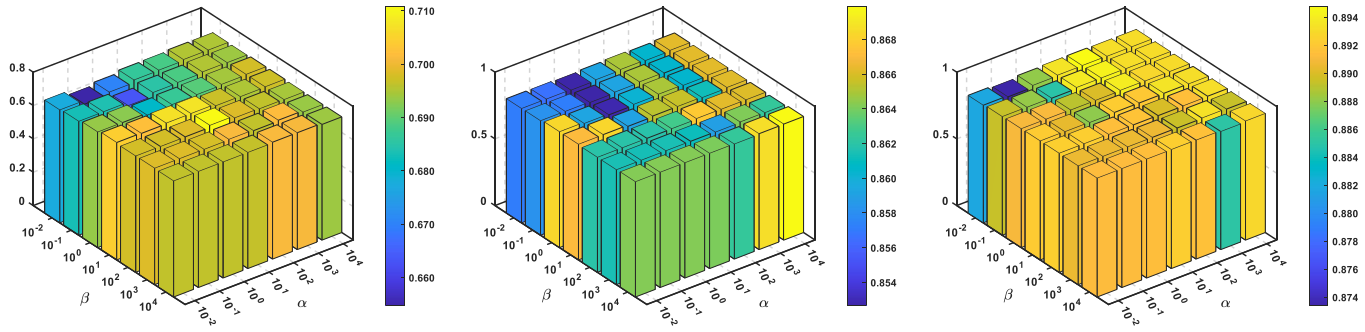
In addition, the average Kappa values over all cases of different BS methods with KNN and SVM classifiers on three datasets are listed in Tables V and VI, respectively. The comparison of average Kappa presents almost the same changes as that of the average OA, which indicates that the proposed method has advantages on three datasets, especially on Indian Pines and Pavia University.

3) Other Discussions: It is found that the proposed method does not present particularly obvious advantages on Salinas as on the other two datasets. This phenomenon occurs not only in this article, but also in others. For example, it was stated that the difference of compared methods is not very obvious on Salinas in [35], and almost all the BS methods give consistently good performances on Salinas in [42]. In this part,

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TABLE VII
Z AND p VALUES WHEN COMPARING SCDBS WITH THE OTHER METHODS

	MVPCA	TRC-OC-FDPC	FNGBS	SimRank	SSR	MDSR	GCSR
Z value	-2.2014	-2.2014	-1.9917	-2.2014	-2.2014	-2.2014	-2.2014
p value	0.0277	0.0277	0.0464	0.0277	0.0277	0.0277	0.0277



(a) (b) (c)

Fig. 5. Average OA values by KNN classifier under different α and β values. OA values for (a) Indian Pines, (b) Pavia University, (c) and Salinas.

the reason for this open question is preliminarily discussed. It is believed that the image of Salinas has six distinctive classes, which have obvious spectral characteristics and can be easily distinguished [13]. Therefore, the classification performance, including OA and Kappa, on this dataset is generally better than that on Indian Pines and Pavia University, and the capabilities of different methods may not be significantly separated on this dataset as well. In addition, according to some research [43], the spectral variability may also be related to the band redundancy and would influence the performance of classifiers, which may be a deep-seated reason and need to be specially studied.

C. Statistic Test

To show that whether the proposed method offers significant differences with the compared methods, the Wilcoxon signed-rank test [44], [45] is conducted between the proposed SCDBS and each of the compared methods. In the test, the averaged OA on each dataset with each classifier is used, meaning that six data items from Tables III and IV are collected to form the representation of each method. Here, the null hypothesis is that the proposed SCDBS has no significant differences with the compared method, and the level of significance considered is $\alpha = 0.05$. When $|Z| > 1.96$ or $p < \alpha$, it is indicated that the test rejects the null hypothesis that there is no difference between the compared two methods at the 5% significance level. The Z values and p values of the Wilcoxon signed-rank test are shown in Table VII, from which it is found all Z values are less than -1.96, and all p values are less than 0.05. Thus, it can be concluded that SCDBS has significant differences with all the compared methods.

D. Parameter Study

To explore the effect of α, β, and p on HSI classification, experiments are carried out on the three datasets with both KNN and SVM classifiers. Since α and β are the trade-off parameters, and p is used to adjust the sparsity level, the following two groups of experiments are conducted.

1) Parameters α and β: In this experiment, the influence of α and β is presented simultaneously. The range of α and β is set to $[10^{-2}, 10^{-1}, \ldots, 10^{3}, 10^{4}]$ under the preset p being 0.5. Under each combination of α and β, the average OA of different numbers of bands (5:5:50) as in Section IV-B2 is used as the indicator of the corresponding performance.

The average OA values by KNN and SVM classifier under different α and β values are shown in Figs. 5 and 6, respectively. The following observations can be found from the six subfigures. 1) For different datasets, the optimal parameter combination to achieve the best classification performance is different, which may be related to the attributes of the dataset, such as the classes of ground objects, and band correlations. 2) The classification performance evolution of SVM and KNN classifiers with respect parameter combinations is not completely consistent, which reflects the different principles of the two kinds of classifiers. However, they are similar in most cases, which reflects the robustness of the selected bands to different classifiers. 3) In general, when α is fixed to different values, the average OA increases gradually, or increases first and then decreases with the increase in β. When α and β are exchanged, the conclusion is similar. 4) After removing the worse parameters (both small α and β, corresponding to the blue area in the figures), the performance variation is not very great, especially on the Pavia University and Salinas datasets.

2) Parameter p: In this experiment, the range of p is set as $[0.25, 0.50, 0.75, 1.00]$ to explore its influence. Under each p value, the range of α and β is the same as the above experiment. Finally, the best average OA among the different combinations of α and β is used as the performance indicator for the proposed method with different p values.

The best average OA values by KNN and SVM classifiers under different p values are shown in Figs. 7 and 8, respectively. It can be observed that KNN and SVM have the similar classification evolution under different p values. More

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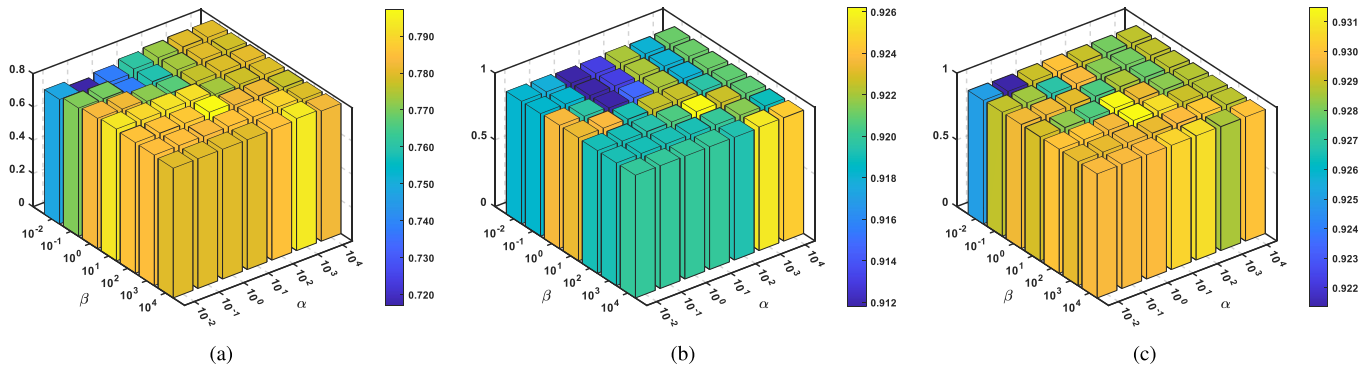


Fig. 6. Average OA values by SVM classifier under different α and β values. OA values for (a) Indian Pines, (b) Pavia University, and (c) Salinas.

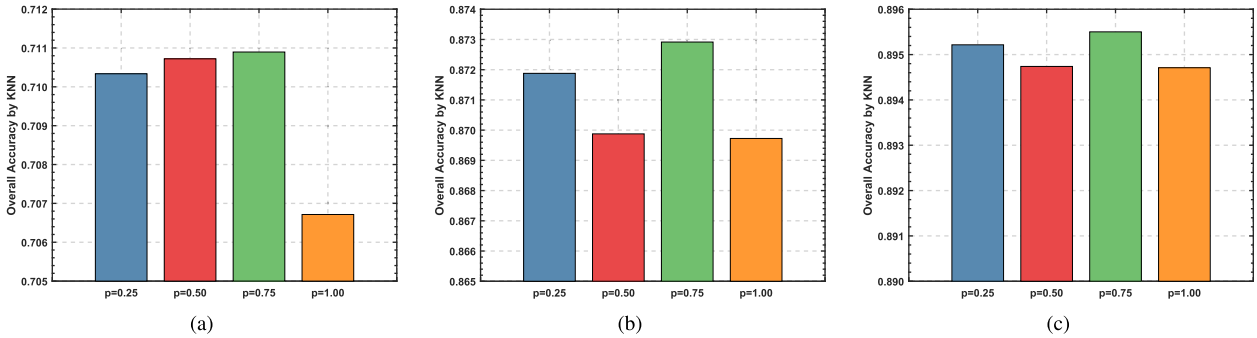


Fig. 7. Best average OA values by KNN classifier under different p values. OA values for (a) Indian Pines, (b) Pavia University, and (c) Salinas.

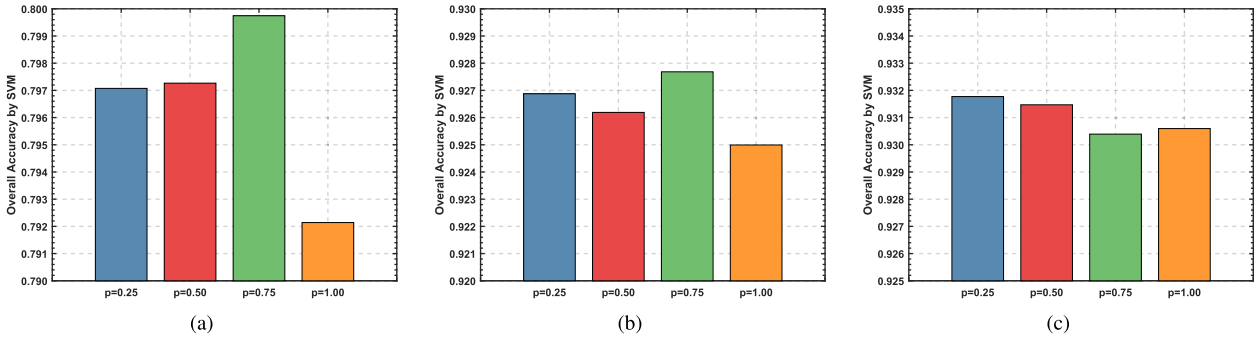


Fig. 8. Best average OA values by SVM classifier under different p values. OA values for (a) Indian Pines, (b) Pavia University, and (c) Salinas.

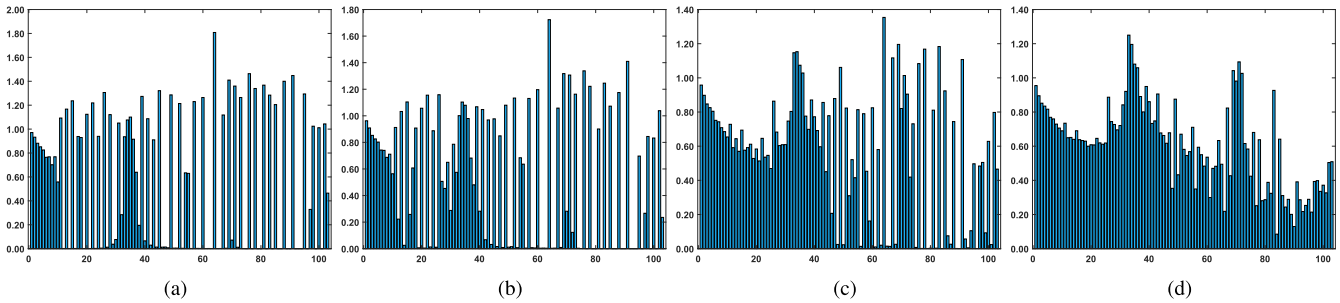
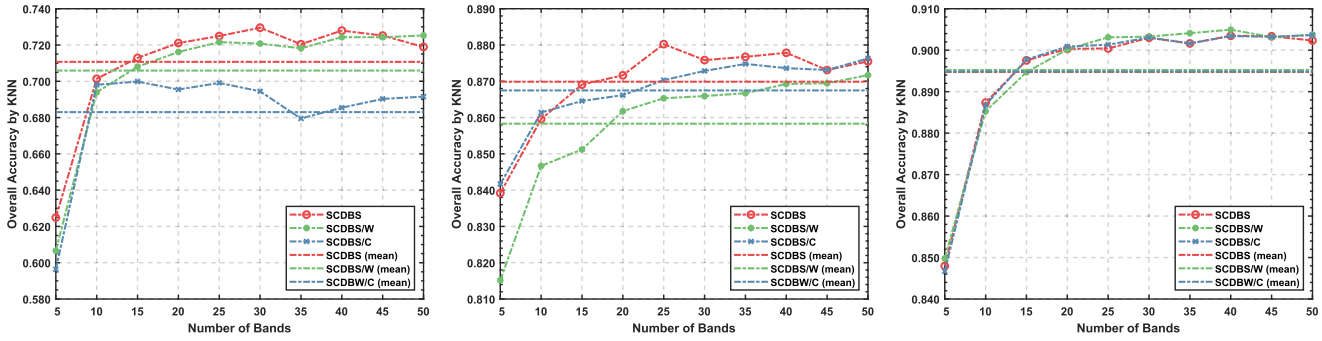


Fig. 9. Sparsity illustration of different p values. (a) $p = 0.25$. (b) $p = 0.50$. (c) $p = 0.75$. (d) $p = 1.00$.

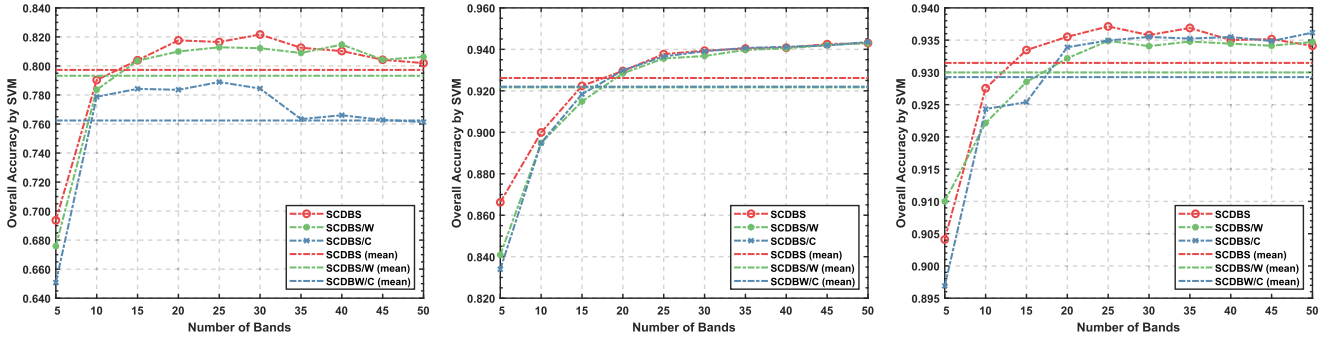
importantly, whether SVM or KNN is used as the classifier, when $p < 1.00$, the classification performance of the proposed method is higher than that of $p = 1.00$. This may because when p is closer to 0, the obtained row sparsity can be more closer to the strict sparsity constraint by the $\ell_{2,0}$ norm, as shown in Fig. 9, where the value of each bar, also the representation weight of each band, is computed as the ℓ_2 norm of each row of the sparse representation matrix $\mathbf{W}$. This will reduce the chance that adjacent bands have a similar level of reconstruction weight. Therefore, the bands selected based on

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(a) (b) (c)

Fig. 10. OA curves and the mean by KNN classifier. OA values for (a) Indian Pines, (b) Pavia University, and (c) Salinas.



(a) (b) (c)

Fig. 11. OA curves and the mean by SVM classifier. OA values for (a) Indian Pines, (b) Pavia University, and (c) Salinas.

TABLE VIII
RUNNING TIME (SECONDS) OF DIFFERENT METHODS WITH 30 BANDS BEING SELECTED

	MVPCA	TRC-OC-FDPC	FNGBS	SimRank	SSR	MDSR	GCSR	SCDBS
Indian Pines	0.0302	0.1322	0.0402	4.0230	1.5242	**0.0191**	1.1072	0.1895
Pavia University	0.1677	0.2158	0.1154	15.7327	6.1891	**0.0105**	3.0995	0.4986
Salinas	0.1013	0.3086	0.1333	9.2032	7.5421	**0.0142**	6.4097	0.8846

the reconstruction weights will be more likely to have lower redundancy, which is beneficial to classification.

E. Ablation Study

Ablation experiments are carried out to verify the effectiveness of the proposed method. Therefore, the following three implementations are designed and compared.

1) *SCDBS/W:* Implemented by replacing correlation-based weighted sparse representation of the proposed SCDBS with nonweighted sparse representation, i.e., setting the diagonal matrix $\mathbf{D}$ in (5) as an identity matrix $\mathbf{I}^{n\times n}$.
2) *SCDBS/C:* Implemented by removing the designed correlation minimization regularization term in (7), i.e., setting the regularization parameter β as 0.
3) *SCDBS:* The proposed complete implementation.

The OA under different numbers of selected bands and the average OA used by previous experiments in Section IV-B are compared, in which p is fixed as 0.5 for all the three implementations, and α and β adopt the best parameters through grid search. The compared classification results of KNN and SVM classifiers on three datasets are shown in Figs. 10 and 11, respectively. It can be observed that in most cases presented, the proposed SCDBS outperforms SCDBS/W and SCDBS/C, especially on Indian Pines and Pavia University datasets, which demonstrates the effectiveness of both weighted sparse representation and correlation minimization regularization in the proposed method. In addition, it can be found that the performance of SCDBS/W is better than that of SCDBS/C in Figs. 10(a), and 11(a) and (c), which indicates that correlation removal is important to BS.

F. Computational Efficiency

To verify the efficiency of the proposed method, the running time of selecting 30 bands by all the compared BS methods on three datasets is recorded, which includes the necessary preprocessing of each method. All the methods are implemented by MATLAB R2021b on the same platform with an Intel Core i7-12700H, and 16-GB RAM.

According to the comparison results shown in Table VIII, it can be found that SimRank takes the longest time, which is

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mainly caused by its requirement for computing the pairwise similarity via SSIM [40]. On the contrary, MDSR takes the least time among all the compared methods. The main reason is that it only randomly samples 50 pixels from all spatial pixels, which greatly reduces the dimension of each band. MVPCA generally takes less time than other compared methods, which is consistent with its low computational complexity approximated as $\mathcal{O}(m^2)$. In addition, the computation time of FNGBS is not much different from that of MVPCA. SSR and GCSR take relatively more time, mainly because they need the clustering algorithm to run on the obtained sparse matrix to determine the final selected bands. It can be found that the running time of the proposed method is moderate among all the methods, which is more than that of MDSR, MVPCA, FNGBS, and TRC-OC-FDPC, but less than that of SimRank, SSR, and GCSR. Therefore, our method is computationally temperate while achieving the improved classification performance.

V. Conclusion

To consider and exploit the correlation among bands, SCDBS for HSIs is proposed in this article. Compared with the conventional SSR-based BS, SCDBS adopts the correlation derived weight to conduct weighted sparse reconstruction to improve the representativeness of the selected bands and also designs a correlation minimization regularization term to reduce the redundancy and improve diversity among the selected bands. In addition, the proposed SCDBS imposes the $\ell_{2,0<p\leq 1}$ norm-based constraint to the sparse coefficients, which extends and unifies the commonly used $\ell_{2,1}$ norm to provide adjustable sparsity level. Moreover, an iterative algorithm with moderate computational complexity is designed to optimize the proposed SCDBS model. Extensive experiments are carried out on three benchmark datasets, and results have demonstrated the effectiveness of each part in the proposed method, and the improved classification performance with acceptable computational cost.

In the current version, the spatial information of HSIs is not considered. In the future work, the simultaneous integration of spectral and spatial relationships for BS is a direction of improvement for this work. In addition, the study of the relationship between spectral variability and band redundancy is also a noteworthy direction for the BS of HSIs.

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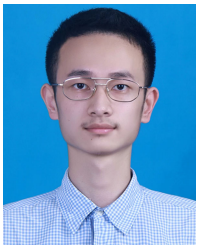
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